

Asset Pricing During Potential Disasters

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1 Introduction

The paper proposes an asset pricing model with a slowly unfolding disaster risk during which disaster-specific news constitute a major source of variation in asset prices often independently from contemporaneous realizations of fundamental variables. Existing models of disaster risk were designed with the purpose of explaining asset pricing moments using the notion of an economic disaster, or a sudden and unexpected jump in aggregate consumption (see e.g. Rietz (1988), Barro (2006), Gabaix (2012) and Nakamura et al. (2013)). These papers argue that standard models in which consumption jumps are not explicitly accounted for understate the importance of tail risk and for this reason fail to explain crucial features of asset pricing dynamics. They focus on how the general possibility of disasters can explain asset pricing phenomena and/or how price behavior differs on average across normal and disaster periods. These models, however, do not explicitly address how asset prices evolve during a specific unfolding potential disaster and might thus not serve best as a systematic framework in which specific disaster episodes can be analyzed. There are three main reasons behind this.

Firstly, in existing models, stock market disaster is like a nuclear strike (or a series of low-yield nuclear strikes) - countries can recover from it over time, but there is no mechanism in the models via which sudden "un-nuking" can take place. This is because in models without learning, price recovery is closely linked to consumption recovery, which is only gradual even in models with sharp but partial post-disaster recoveries (see e.g. Nakamura et al. (2013)). Stock markets recover only gradually in models with learning as well, because good news in such models is a persistent absence of

signals about disasters, which needs several periods of "tranquility" to induce substantial downward shifts in the perceived likelihood of disasters.¹

Yet, stock market recovery during covid substantially preceded economic recovery and it did so largely as a result of the arrival of good news (Fed intervention, government relief packages, vaccine inventions) that caused sharp upwards jumps in growth expectations, synonymous with "un-nuking" in the financial markets. As November 2020 article in WSJ cites Seema Shah saying, "2020 is a year we won't ever forget and yet somehow equity markets have completely erased it from their memory". In fact, when the financial markets recovered, the global economy was still in disaster regime.

Secondly, in existing models, stock market disasters can occur only in response to macroeconomic disasters, yet this need not always be the case. Barro and Ursua (2017) analyze long-term data for 30 countries to find that conditional on a market crash, the probability of a minor depression is 31% and of an economic disaster - only 10%. During the Black Monday of October 1987, the stock market experienced a 26% crash, the largest single-day fall in the stock market history, yet it did not take place as a result of a macroeconomic disaster, nor did it cause one. In fact, the market recovered 57% of its losses in just two trading days largely as a result of the Fed's intervention. Similar stock market swings could also take place in response to evolving climate change threats and independently from macroeconomic disasters, as a result of market responding to potential global supply chain disruptions from climate change threats that do not materialize or are at least partially resolved unexpectedly or as a result of interventions.

Thirdly, existing models assume that investors either know disaster probabilities each period or observe other parameters of the fundamental process governing disasters, such as disaster probabilities in low and high risk states or regime Markov-switching probabilities. These parameters are extremely hard to calibrate in general due to the rarity of disasters, and more so in the context of a specific unfolding disaster, for which historical data could be even less relevant, especially if the source of the disaster is an unprecedented event like covid.

All three issues are addressed by the model set forth in this paper. In the model,

¹It is natural that general models of disaster risk do not feature good news that might induce downward jumps in disaster probabilities. After all, this would amount to the assumption that during normal times investors can observe signals that suggest that already very unlikely event of a general economic disaster has suddenly become even much less likely. During an unfolding potential disaster, however, good news is a sensible (and crucial) concept, since it facilitates sudden arrival of positive information about the future evolution of a specific unfolding disaster.

investors can observe both good and bad news of varying intensity, which can cause jumps in prices in either direction and independently from consumption realizations. Hence, stock prices can recover faster than consumption and can experience brief disasters without macroeconomic disasters. The problem of calibration is also simplified since disaster probabilities are pinned down endogenously using the rational inattention theory, which is a key facilitator of all three features.

To illustrate these aspects more clearly, let R and s denote, respectively, a binary regime indicator and a binary noisy signal about it. Let R_1 denote a good regime (in which consumption growth probability is high) and R_2 - a disaster regime (in which consumption growth probability is low). Let s_i denote a signal realization that suggests that true regime is R_i , $i = 1, 2$. Hence, s_1 corresponds to good news, while s_2 - to bad news. How informative each realization is depends on the unit cost of information λ , which proxies fundamental disaster-induced uncertainty: when s_1 is observed along with λ remaining high, good news arrives but it does not affect disaster-induced uncertainty much and is only modestly informative about the future path of disaster evolution (e.g. Moderna and Pfizer vaccines have passed Phase II trials, but since Phase III is still awaiting, the good news does not reduce covid-induced uncertainty very much). When s_1 is observed along with a downward jump in λ , good news arrives that also substantially reduces disaster-induced uncertainty (e.g. the Fed announces covid intervention programs that reduce uncertainty about the economic impact of the covid pandemic). Likewise, a pair of s_2 and high λ could correspond to early days of covid during which the properties of the virus were not well-understood, while a pair of s_2 and low λ could correspond to the times when the WHO announced the covid as a global pandemic. Hence, λ can also be viewed as a proxy of news intensity (or how high a signal realization is on good or bad news spectrum). As will be shown shortly, the effects of the realizations of s and λ are parsimoniously passed down to prices, with the directions of shifts depending on whether news is good or bad, and the magnitude - on how small realized λ is.

Mathematically, signal informativeness is described by $F(s_i|R_i)$, $i = 1, 2$, where $F(\cdot)$ is a joint distribution of s and R , and a function of λ . $F(\cdot)$ is not exogenously specified, but induced by the information choice problem of the representative agent, in which he tries to select $F(\cdot)$ in a way that maximizes its ex-ante usefulness net costs of learning, which is a linear function of λ . The information choice problem is a rule-of-thumb description of the information acquisition process of investors and is to be understood as a consequence of sophisticated trial and error by financial institutions

with substantial data-processing capacity and experienced management. λ thus is not a quantity that is assumed to be explicitly observed and taken into account, but it reflects the effort financial institutions implicitly face during information acquisition and processing, and can potentially be estimated using textual analysis of newspaper articles.²

Unlike other disaster models that rely on the specification of true arrival rate of disaster jumps and/or regime-switching probabilities, the RI framework of this paper provides an intuitive theory of how market beliefs are determined without relying on true probabilities, and in doing so greatly simplifies the problem of calibration. The theory states that market beliefs about the possibility of a disaster are pinned down by three interacting forces - the representative investor trying to gather information from most informative sources (or trying to select s with as high $F(s_i|R_i)$ for both i as possible), how high the unit cost of information λ is (or how high the fundamental uncertainty is) and what kind of news arrives (whether s_1 or s_2 is observed). The true stochastic process governing disasters then enters the model implicitly through λ and signal realizations, and is not explicitly modeled, since investors are not assumed to know it anyway.

One of the main implications of the model is that time- t equilibrium price $P_{Z,t}$ of an asset with pay-off function Z following a signal realization s_k is given by

$$P_{Z,t} = F_{\lambda_t}(R_i|s_k) P_{Z,t}^i + F_{\lambda_t}(R_{-i}|s_k) P_{Z,t}^{-i},$$

where $P_{Z,t}^i$ is the time- t price of the asset in regime R_i if it were fully observable (i.e. $P_{Z,t}^i$ is a price on a time- t node of a binomial tree with the growth probability corresponding to R_i) and $F_{\lambda_t}(R_i|s_k)$ is the posterior belief of the representative investor about R_i after observing s_k in period t under optimal information strategy and cost of information λ_t .³ Except a few special cases (see discussion under Theorem 4.1),

²Contrary to other models of Rational Inattention in which agents learn about several variables and try to optimally allocate attention across them, there is a single subject of learning in this model - an unfolding potential disaster and its effects. Attention allocation across different variables is not explicitly modeled but is ingrained in the information choice problem of the representative investor: the agent wants to select the most useful information-gathering strategy possible, yet informational sources are plenty, while attention is scarce and learning - effortful. The agent then tries to gather information from sources that he deems most useful and by doing so implicitly maximizes $F(s_i|R_i)$ within his capacity for learning.

³ s_k does not have a time subscript, because the market can observe s_1 or s_2 in each period. The posterior regime probabilities for a given signal realization do not depend on the history of signals, but the latter affects the probability of good news arrival in each period. Along with changes in the

$F_{\lambda_t}(R_i|s_i) > F_{\lambda_t}(R_{-i}|s_i)$ and the difference is larger the smaller λ_t is, with $F_{\lambda_t}(R_i|s_i)$ always converging to 1 as λ_t shrinks to zero. Hence, changes in s and λ can cause sharp shifts in the weights the two regime prices receive and thus cause jumps in the observed price itself. These jumps can happen independently from consumption realizations, albeit the jumps would be largest when bad news arrives along with negative consumption outcomes. These two effects are separate in the expression for $P_{Z,t}$: consumption realizations are reflected in regime prices only, while news arrival in the weights they receive. These two events, however, are not statistically independent: bad consumption realizations at the end of a period increase the probability of bad news in the next period and vice versa.

To illustrate how this happens, suppose $i = 1$ and the disaster episode corresponds to the covid pandemic. Let $t = -1$ denote the period right before the covid outbreak and note that $P_{Z,-1} = P_{Z,-1}^1$, i.e. the market was oblivious to the covid risks before it emerged.⁴ At time zero, the possibility of covid emerges (non-zero λ_0 appears and s_2 is observed) and the observed price jumps to

$$P_{Z,0} = F_{\lambda_0}(R_1|s_2)P_{Z,0}^1 + F_{\lambda_0}(R_2|s_2)P_{Z,0}^2.$$

Since in its early days covid had a negligible economic impact, if any, $P_{Z,0}^i > P_{Z,-1}^i$ for both i . However, since $F_{\lambda_t}(R_2|s_2) > F_{\lambda_t}(R_1|s_2)$, $P_{Z,0}^2$ receives more than half of the weight in determining $P_{Z,0}$ and so the latter jumps down relative to $P_{Z,-1}$, yet modestly because of λ_0 being high, which reflects the lack of understanding of the true nature of covid. In subsequent periods, bad news keeps arriving that also sheds light on the nature of covid, which is reflected in gradual modest downward jumps in λ_t . Jumps in λ_t in turn cause jumps in $F_{\lambda_t}(R_2|s_2)$ and hence in $P_{Z,t}$. Eventually, as the WHO announces the covid as a global pandemic (i.e. λ_t jumps down sharply and s_2 is observed) and governments impose lockdowns ($P_{Z,t}^i$ goes down for both i)⁵, $P_{Z,0}$ crashes.

Other models of general disaster risk with learning could probably yield the same insights as above when applied to the covid pandemic. However, most of the models

unit cost of information, this is what facilitates jumps in posterior regime beliefs mathematically, since belief updating in response to news is typically incremental in settings in which beliefs are formed as a result of Bayesian updating based on the whole history of signals.

⁴Note that this does not require that the market was oblivious to *any* disaster risk at time -1, since the model can be extended so that regime prices incorporate general disaster risk.

⁵Since lockdowns have resulted in aggregate consumption declines, both regime prices have experienced down-moves along their binomial trees.

with learning can be solved only in semi-analytic form and the full numerical solution requires the calibration of parameters that would not be feasible in the context of a specific disaster. At the same time, what general models of disaster risk would have difficulty explaining is the sudden recovery from covid, which can be illustrated intuitively in the framework of this paper as follows: after the stock market crash, the US government and the Fed announce a series of relief packages and measures, and vaccine news gradually arrive, which constitute further downwards jumps in λ_t and the arrival of good signal realizations.⁶ This results in a series of jumps in the weight of $P_{Z,t}^1$, initially because a good signal realization shifts most of the weight in the convex combination towards $P_{Z,t}^1$ and then as a result of gradual yet sharp increases in $F_{\lambda_t}(R_1|s_1)$ due to corresponding declines in λ_t over a few periods. At the same time, the projections of sharp U-turn economic recovery become prevalent, which boost growth expectations in regime R_1 and further increase $P_{Z,t}$ through $P_{Z,t}^1$. Occasional modest and temporary negative swings in the recovery process can be understood in a similar spirit based on disaster-specific news arrival or changes in regime prices.

The separation of news effects and consumption growth effects on asset prices is also the main mechanism that facilitates stock market disasters without macroeconomic disasters: the prices can crash in the absence of persistent declines in consumption, or in extreme cases - in the absence of any consumption declines at all. These features are much harder to generate in models of general disaster risk.

Lastly, although the framework in this paper is posed in the language of disasters, it could be used to model any major source of uncertainty that affects asset prices. At the same time, it sets forth a novel application of rational inattention theory in asset pricing, which has so far been constrained to two-period models with Normally distributed signals. The framework of this paper builds novel theoretical tools for modeling learning based on the discrete choice framework of Matejka and McKay (2015), which could potentially be applied to a large class of models. Hence, the contribution of this paper is also theoretical.

⁶The downward jumps in λ_t so far have been reflecting the arrival of news illuminating the nature of covid. Once covid was adequately understood and the outbreak was declared as a global pandemic, substantial uncertainty still remained regarding the severity of the economic impact. The Fed and government interventions substantially decreased that uncertainty, which reflects further downward jumps in λ_t . The same points apply to vaccines arrival.

2 Model Description

The economy is inhabited by a continuum of identical agents with time-separable log-preferences who live for T periods. In each period, the economy is in two possible regimes - a Good regime and a Disaster regime. R denotes a binary regime indicator with $R = G$ indicating that the current regime is Good and $R = D$ indicating that the current regime is Disaster. In each regime, the uncertainty evolves along a binomial tree and x_t^j - j -th node in period t - is associated with aggregate consumption $\omega(x_t^j)$. In the Good regime, the probability of an up-move (aggregate consumption growth) is $p_G > 1/2$, while in the Disaster regime - $p_D < 1/2$. Hence, the two regimes yield the same outcome sets in each period but induce different probability distributions over them. Lower the p_D , the more the probability is concentrated around bad outcomes in each period in the Disaster regime. Hence, p_D can be thought of as a parameter that controls disaster intensity. The markets are complete in the sense that there exists an Arrow-Debreu security for each node x_t^j . The current regime is unobservable and the representative investor wants to i) select an optimal information-gathering strategy to learn about the current regime and ii) choose a portfolio of Arrow-Debreu securities that maximizes his T -horizon expected utility w.r.t beliefs induced by his information.

Any information strategy is defined by the joint distribution of R and signal function s_λ . s_λ is a noisy binary function that indicates the regime the economy is - a realization s_G of s_λ suggests the economy is in the Good regime and a realization s_D of s_λ suggests the economy is in the Disaster regime. p_0 is the prior belief of the representative agent about the likelihood of being in the Good regime at time 0. The unit cost of information parameter λ controls the effort (or induced disutility) that is required to reduce regime uncertainty by one unit. Intuitively, the optimal information strategy selection is described as follows: the representative investor wants to select a signal function s_λ such that if he were to optimize his portfolio as if the signal realizations were error-free⁷, no other signal function would yield higher ex-ante expected utility. The choice variable in the information selection problem is the joint distribution of s_λ and R . Hence, the set of signals to choose from is not exogenously restricted and the representative investor has access to any information strategy he deems feasible given

⁷Let Good portfolio denote the portfolio of Arrow-Debreu securities the investor would select in the Good regime and define Disaster portfolio analogously. Optimizing as if signal realizations were error-free corresponds to choosing Good portfolio after observing s_G and choosing Disaster portfolio after observing s_D . As will be shown later, the optimal portfolio induced by the posterior beliefs is a convex combination of the Good portfolio and a Disaster portfolio.

the unit cost of information λ (see Section 3 for mathematical details). Denote the optimal signal function by s_λ henceforth.

Given the optimal information strategy, the representative investor optimizes his portfolio using the posterior beliefs $P(x_t^j | s_\lambda = s)$ induced by Bayesian updating and a realization s of the optimal signal function s_λ , i.e. the investor solves

$$\max_{c(x_t)} \sum_{t=0}^T \beta^t \sum_{j=1}^{t+1} \log(c(x_t^j)) P(x_t^j | s_\lambda = s)$$

subject to

$$\sum_{t=0}^T \sum_{j=1}^{t+1} q(x_t^j) c(x_t^j) \leq \sum_{t=0}^T \sum_{j=1}^{t+1} q(x_t^j) \omega(x_t^j),$$

where $c(x_t) = \left((c(x_t^j))_{j=1}^{t+1} \right)_{t=0}^T$ and $q(x_t^j)$ and $c(x_t^j)$ denote, respectively, the state-price and Arrow-Debreu security corresponding to x_t^j .

The sequence of events is as follows: the representative agent starts with the prior belief p_0 about being in the Good regime, selects optimal signal s_λ , observes its realization s , does Bayesian updating of his beliefs about the current regime and consequently, of posteriors $P(x_t^j | s_\lambda = s) \forall x_t^j$. He then chooses the optimal portfolio induced by the posterior beliefs and prices all Arrow-Debreu securities. At time 1, $\omega(x_1)$ is realized, and based on the realization, the agent does Bayesian updating of his time-zero posterior belief about being in a Good regime. Call the resulting belief p_1 and note that it is time-1 analogue of p_0 . From this point, the time-1 sequence of events is analogous to the time-0 sequence and the process repeats until time T .

The model is formulated in terms of Arrow-Debreu securities for convenience. The price of any asset with a known pay-off function can be readily recovered from state prices using the no-arbitrage argument. Although λ has been introduced as a constant parameter, it can be time-dependent and evolve deterministically or stochastically without affecting model tractability. This is because the cost of information is an exogenous object to the investor who optimizes information strategy every period.

3 Precise Formulation of the Signal Selection Problem

The formulation of the signal selection problem is inspired by the Rational Inattention (RI) discrete choice framework of Matejka and McKay (2015). See the Appendix for a brief summary of the RI framework of Matejka and McKay (2015) and for the mapping between their framework and the signal selection problem that follows.

Let $\mathcal{P}_G$ ($\mathcal{P}_D$) denote the portfolio of Arrow-Debreu securities that solves the portfolio selection problem of the representative investor in the Good (Disaster) Regime if it were fully observable, and let $\mathcal{P}_s$ denote a random variable that would equal $\mathcal{P}_G$ if the good signal were observed and would equal $\mathcal{P}_D$ otherwise. $V(\mathcal{P}_s, R)$ denotes the expected utility of $\mathcal{P}_s$ in Regime R , and $G(R)$ - the prior beliefs of the representative investor about being in either regime, i.e. $G(G) = p_0$ and $G(D) = 1 - p_0$. $F(s, R)$ is the joint distribution function of signal function s_λ and regime indicator R , while $F(s|R)$ is the distribution function of s_λ conditional on R . The information choice problem is defined as follows: the representative investor wants to

$$\max_{((F(s|R))_s)_R} \left\{ \sum_{R=G,D} \sum_{s=s_G, s_D} V(\mathcal{P}_s, R) F(s|R) G(R) - c(F) \right\} \quad (3.1)$$

subject to

$$\begin{aligned} \forall s: \quad & F(s|R) \geq 0 \quad R = G, D \\ & \sum_s F(s|R) = 1, \quad R = G, D. \end{aligned}$$

$\sum_R \sum_s V(\mathcal{P}_s, R) F(s|R) G(R) \equiv U(F)$ is a utility-based measure of signal usefulness. To see this, note that with $R = G$, $V(\mathcal{P}_G, G) > V(\mathcal{P}_D, G)$ and with $R = D$, $V(\mathcal{P}_D, D) > V(\mathcal{P}_G, D)$. Hence, $U(F)$ is maximized when $F(s_G|G) = F(s_D|D) = 1$ and strictly decreases monotonically as the two quantities move away from 1 towards zero.

Since $F(s|R)G(R) = F(s, R)$ by Bayes Law, any selection of $((F(s|R))_s)_R$ pins down a joint distribution of s and R . Conversely, since any joint distribution pins down conditional distributions, it follows that the optimal choice of $((F(s|R))_s)_R$ leads to the optimal choice of $F(s, R)$. Hence, the signal selection problem can be interpreted either in terms of choosing the joint distribution $F(s, R)$ or in terms of selecting conditional probabilities $((F(s|R))_s)_R$.

$c(F)$ is the cost of information strategy and is modeled as

$$c(F) = \lambda (H(G) - E_s[H(F(.|s))]). \quad (3.2)$$

$H(G)$, the entropy of $G(R)$, measures the uncertainty inherent in prior beliefs, while $H(F(.|s))$, the entropy of $F(R|s)$ - the uncertainty inherent in the posterior beliefs after observing a signal realization s . $E_s[H(F(.|s))]$ measures the average entropy of posterior beliefs across signal realizations and hence $H(G) - E_s[H(F(.|s))]$ is the average reduction in the uncertainty about R as a result of knowing s . Non-negative constant λ is the cost of uncertainty reduction by one unit.

The signal selection problem has been formulated in a static setting for tractability. To generalize it to an arbitrary period t , one only needs to replace R , $\mathcal{P}_G$, $\mathcal{P}_D$, $\mathcal{P}_s$, $V(\mathcal{P}_s, R)$, $G(R)$ and λ with R_t , $\mathcal{P}_{G,t}$, $\mathcal{P}_{D,t}$, $\mathcal{P}_{s_t}$, $V_t(\mathcal{P}_{s_t}, R_t)$, $G_t(R_t)$ and λ_t , respectively. R_t denotes time- t regime indicator, while $\mathcal{P}_{G,t}$ ($\mathcal{P}_{D,t}$) is the portfolio of Arrow-Debreu securities that solves the time- t portfolio selection problem⁸ of the representative investor in the Good (Disaster) Regime if it were fully observable. $\mathcal{P}_{s_t}$ is a random variable that would equal $\mathcal{P}_{G,t}$ if the good signal were observed at time t and would equal $\mathcal{P}_{D,t}$ otherwise. $V_t(\mathcal{P}_{s_t}, R_t)$ is the expected utility of portfolio $\mathcal{P}_{s_t}$ in regime R_t at time- t -observable node x_t . $G_t(R_t)$ denotes the time- t prior beliefs about R_t and λ_t is exogenously determined time- t unit cost of information.

4 Optimal Information Strategy

Relative entropy, or Kullback-Leibler (KL) divergence, is a central concept within the context of optimal information strategy. KL divergence of distribution P from distribution Q , denoted by $KL(P||Q)$, is a well-known information-theoretic object that measures how much one distribution differs from another, assuming the two have the same support. Mathematically, letting x be an arbitrary element in the common support of P and Q , $KL(P||Q)$ is defined as

$$KL(P||Q) = E_P \left[\log \frac{P(x)}{Q(x)} \right],$$

⁸Time- t regime-specific portfolio selection problem is merely the problem the representative investor would solve at time- t -observable node x_t in each regime.

i.e. it is a P -expected log-difference between the probabilities induced by the two distributions. $KL(P||Q)$ is always non-negative and equals zero iff $P = Q$. Intuitively, $KL(P||Q)$ measures the expected excess surprise induced by the usage of Q instead of P when the true distribution is P . For this reason, KL divergence is not necessarily symmetric, i.e. in general $KL(P||Q) \neq KL(Q||P)$.

Let R_1 denote the Good regime and note that in the context of this paper, $KL(P(x_t|R_i)||P(x_t|R_{-i}))$ denotes the KL divergence of the probability distribution over x_t induced by regime R_i from the distribution induced by regime R_{-i} . Hence, $KL(P(x_t|R_i)||P(x_t|R_{-i}))$ provides a measure of how much regime uncertainty today affects the likelihood of t -period outcomes: when the KL divergence is close to zero, the two regimes provide similar outlooks about x_t and not knowing which regime is underway does not create a possibility of major mistakes. When $KL(P(x_t|R_i)||P(x_t|R_{-i}))$ is high, however, the outlooks provided by the two regimes differ substantially and choice behavior that is optimal for one regime might carry significant opportunity costs in another.

If we scale $KL(P(x_t|R_i)||P(x_t|R_{-i}))$ by β^t and sum up all scaled t -period relative entropies, we get a time-adjusted object that measures regime-induced divergence in beliefs not only across time- t outcomes, but over the entire T -period tree. Let such object be denoted by $\tilde{KL}(R_i||R_{-i})$, i.e. set

$$\tilde{KL}(R_i||R_{-i}) = \sum_{t=1}^T \beta^t KL(P(x_t|R_i)||P(x_t|R_{-i})).$$

In the context of our model, $\tilde{KL}(R_i||R_{-i})$ has a sharper interpretation. It is the expected utility cost of optimizing the portfolio for regime R_{-i} when the true regime is R_i . Mathematically, letting $\mathcal{P}_i$ denote the optimal portfolio corresponding to R_i , we have

$$\tilde{KL}(R_i||R_{-i}) = V(\mathcal{P}_i, R_i) - V(\mathcal{P}_{-i}, R_i),$$

i.e. $\tilde{KL}(R_i||R_{-i})$ is the foregone amount of expected utility as a result of mistaking R_i for R_{-i} (see (6.9) in the proof of Lemma 6.1 in the Appendix). Thus, $\tilde{KL}(R_i||R_{-i})$ can be interpreted as the cost of false positive about R_{-i} , or the cost of false negative about R_i . For $i = 2$, for instance, $\tilde{KL}(R_i||R_{-i})$ measures the cost of missing a disaster regime.

Henceforth, let F denote the joint distribution of R and s_λ induced by the optimal information strategy. The optimal information strategy is described in the Theorem that follows:

Theorem 4.1. *The following are true under the optimal information strategy,⁹*

i) *The unconditional probability of observing s_1 is given by*

$$p_{s_\lambda} = p_0 \frac{1}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}} + (1 - p_0) \left(1 - \frac{1}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}} \right),$$

ii) *The posterior probability of R_i given $s_\lambda = s_i$ is given by*

$$F(R_i|s_i) = \frac{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_{-i}||R_i)}}{1 - e^{-\frac{1}{\lambda} (\tilde{K}L(R_2||R_1) + \tilde{K}L(R_1||R_2))}} \quad i = 1, 2$$

iii) *For each i , the responses of $F(R_i|s_i)$ to local changes in λ is described as follows:*

$$\frac{\partial}{\partial \lambda} (F(R_i|s_i)) > 0$$

iff

$$\tilde{K}L(R_i||R_{-i}) F(R_i, s_{-i}) > \tilde{K}L(R_{-i}||R_i) F(R_{-i}, s_{-i}),$$

iv) *The optimal information strategy is perfectly informative in the limit,¹⁰ i.e. as $\lambda \rightarrow 0$,*

$$p_{s_\lambda} \rightarrow p_0,$$

$$F(R_i|s_i), F(s_i|R_i) \rightarrow 1,$$

$$P(x_t^j|s_i) \rightarrow P(x_t^j|R_i),$$

v) *The posterior probability of x_t^j given $s_\lambda = s_i$ is given by*

$$P(x_t^j|s_i) = P(x_t^j|R_i) F(R_i|s_i) + P(x_t^j|R_{-i}) F(R_{-i}|s_i), \quad i = 1, 2.$$

Proof. See the Appendix. □

Remark 4.1. As will be shown below, the information strategy has several intuitive yet sophisticated features, which, taken together, could seem to imply unrealistically high level of rationality on the part of representative investor. Such interpretation, however, would not be valid, since these features are not assumptions about the choice

⁹For high values of λ , the agent might find it optimal not to gather any information, i.e. he might select completely uninformative signal function that always returns one value. To keep fidelity with reality, I assume throughout the paper that the prior beliefs, the costs of mistakes and the value of λ are such that the unconditional distribution of s_λ is non-degenerate.

¹⁰The claim rests on the assumption that the prior beliefs about R are unbiased.

of information strategy, but rather implications that follow from plausible axioms (see Section 3 for the description of the signal selection problem). These axioms are not very demanding. They only require that the representative investor have reasonable prior beliefs and actively try to select the most informative strategy he can in the sense of Section 3. He is not required to explicitly know λ and the cost function or be consciously selecting the conditional probabilities, since any chosen information acquisition and processing strategy implicitly induces a joint distribution between signals and regimes and the optimal joint distribution can be interpreted as the beliefs that result from consistent and sophisticated pursuit of information edge.

Such interpretation can be justified using three quick arguments. First, the representative investor is not a single individual but a marginal trader - an amalgam of different institutions that have substantial information processing capacity and are run by managers with years of experience. Second, there are in general very strong incentives to seek information edge in the finance sector and third, the subject of learning - the source of potential disasters - naturally attracts a lot of attention and interest by financial institutions, since it constitutes a major threat to business operations.

Another justification can be found in the analogy with the butterfly effect. When a butterfly swings its wings, the chain of reaction that it induces in the surroundings (including the interactions between the wings and air particles and the chain reactions that they cause) could potentially be described by complex and insightful mathematical expressions. Yet, the butterfly is, of course, completely oblivious to the by-product of its movement and its only motive is to get from Point A to Point B. Similar reasoning applies to the information strategy of the representative investor, albeit with a slight modification: although the information strategy has a rich set of features, they are not *necessarily* consciously selected and are a by-product of continuous and advanced trial and error by sophisticated investors. Moreover, the information choice problem is in general amenable to behavioral modifications, such as the introduction of prior biases or the distortions of Bayesian updating.

Optimal information strategy has a number of intuitive properties that are passed down to equilibrium prices through the posterior beliefs of the representative investor. Posterior regime probabilities induced by the optimal signal function are thus crucial objects in our model. Their nature depends on how much the agent can learn and what kinds of mistakes he wants to minimize. The learning capacity is determined by the unit cost of information λ , while the relative importance of mistakes is governed by

their relative costs. More specifically, the agent can make two kind of mistakes: he might make a false negative judgement about R_i (i.e. observe s_{-i} in regime R_i), or make a false positive judgement about R_i (i.e. observe s_i in regime R_{-i}). The cost of the first kind of mistake is $\tilde{K}L(R_i||R_{-i})$, while of the second kind - $\tilde{K}L(R_{-i}||R_i)$. Parts i)-iv) in the Theorem above describe the roles of the unit cost of information and relative costs of mistakes in the determination of beliefs and signal probabilities, while Part v) indicates how the features of the information strategy are passed down to the beliefs about consumption growth outcomes.

To see how λ and the costs of mistakes interact in the determination of the regime beliefs, it would be useful to compare $F(R_i|s_i)$ against both $F(R_{-i}|s_{-i})$ and $F(R_{-i}|s_i)$. Note first that since $\tilde{K}L(R_2||R_1), \tilde{K}L(R_1||R_2) > 0$, $F(R_i|s_i)$ is always less than 1 for non-zero λ . Whether $F(R_i|s_i) > F(R_{-i}|s_i)$, however, depends both on λ and $\tilde{K}L(R_{-i}||R_i)$, $i = 1, 2$. The condition that $F(R_i|s_i) > F(R_{-i}|s_i)$ can equivalently be expressed as

$$1 - e^{-\frac{1}{\lambda}\tilde{K}L(R_{-i}||R_i)} > e^{-\frac{1}{\lambda}\tilde{K}L(R_{-i}||R_i)} \left(1 - e^{-\frac{1}{\lambda}\tilde{K}L(R_i||R_{-i})}\right), \quad (4.1)$$

which indicates that whenever λ is small enough, $F(R_i|s_i) > F(R_{-i}|s_i)$ for all values of $\tilde{K}L(R_2||R_1)$ and $\tilde{K}L(R_1||R_2)$ (see Part iv) in the Theorem above). Intuitively, when the cost of information is small, the optimal information strategy is very informative and $F(R_i|s_i)$ is much higher than $F(R_{-i}|s_i)$ irrespective of the relative costs of the two kinds of mistakes. Whenever λ is not small, this need not be the case, however, and whether $F(R_i|s_i) > F(R_{-i}|s_i)$ depends on the relative cost $\frac{\tilde{K}L(R_i||R_{-i})}{\tilde{K}L(R_{-i}||R_i)}$. Whenever $\frac{\tilde{K}L(R_i||R_{-i})}{\tilde{K}L(R_{-i}||R_i)} \leq 1$ (equivalently, when the cost of mistaking R_i for R_{-i} is weakly smaller than the cost of mistaking R_{-i} for R_i), the posterior probability of R_i still receives higher weight following s_i and for all values of λ than does the posterior probability of R_{-i} .¹¹ This is because $F(R_{-i}|s_i)$ is directly proportional to $F(R_{-i}, s_i)$ - the probability of a false positive about R_i - which is a weakly costlier mistake by $\frac{\tilde{K}L(R_i||R_{-i})}{\tilde{K}L(R_{-i}||R_i)} \leq 1$. Hence, $F(R_i|s_i)$ being higher than $F(R_{-i}|s_i)$ for all values of λ reflects the preferences of the optimizing agent to minimize costlier mistakes.

Whenever $\frac{\tilde{K}L(R_i||R_{-i})}{\tilde{K}L(R_{-i}||R_i)} > 1$, however, and the unit costs of learning are sizable, condition (4.1) might not hold any more and the posterior probability of R_i might be smaller

¹¹Note that $e^{-\frac{1}{\lambda}\tilde{K}L(R_{-i}||R_i)} < 1$ always holds and whenever $\tilde{K}L(R_i||R_{-i}) \leq \tilde{K}L(R_{-i}||R_i)$, $1 - e^{-\frac{1}{\lambda}\tilde{K}L(R_{-i}||R_i)} > 1 - e^{-\frac{1}{\lambda}\tilde{K}L(R_i||R_{-i})}$.

than the posterior probability of R_{-i} even after observing s_i .¹² When that is the case, it is always true that $F(R_{-i}|s_{-i}) > F(R_i|s_{-i})$.¹³ Intuitively, the optimizing agent faces a trade-off between higher $F(R_i|s_i)$ and higher $F(R_{-i}|s_{-i})$. Increasing $F(R_i|s_i)$ comes at the cost of accepting relatively smaller $F(R_{-i}|s_{-i})$, which is inversely related to the probability of the costlier mistake. Hence, the optimizing agent prefers to err on the side of safety, and when the stock of learning capacity is small enough (λ is high enough), adequate reduction of costlier mistakes requires (or makes it optimal) that $F(R_i|s_i)$ be less than $F(R_{-i}|s_i)$. In the language of disasters, whenever understanding the driver of the potential disaster is difficult and missing out a disaster is much costlier than falsely suspecting it, the optimal information strategy dictates that the most effort be allocated towards ensuring that the probability of missing the signs of an impending disaster is small.

The motive of minimizing the costlier mistake is also evident when comparing $F(R_i|s_i)$ against $F(R_{-i}|s_{-i})$, and inspecting p_{s_λ} . Whenever $\frac{\tilde{K}L(R_i||R_{-i})}{\tilde{K}L(R_{-i}||R_i)} = 1$, $F(R_i|s_i) = F(R_{-i}|s_{-i})$ and $F(R_i|s_i) > F(R_{-i}|s_{-i})$ iff $\frac{\tilde{K}L(R_i||R_{-i})}{\tilde{K}L(R_{-i}||R_i)} < 1$. Likewise, p_{s_λ} is decreasing in $\tilde{K}L(R_2||R_1)$ and increasing in $\tilde{K}L(R_1||R_2)$. All these features are straightforward to see by inspection.

The expected utility costs of mistakes also dictate how posterior regime probabilities respond to local changes in λ under the optimal information strategy. By Part iii) of Theorem 4.1, $\frac{\partial}{\partial \lambda}(F(R_i|s_i)) > 0$ iff

$$\tilde{K}L(R_i||R_{-i})F(R_i, s_{-i}) > \tilde{K}L(R_{-i}||R_i)F(R_{-i}, s_{-i}). \quad (4.2)$$

Whenever the stakes of a mistake are equally high in the two regimes ($\tilde{K}L(R_i||R_{-i}) = \tilde{K}L(R_{-i}||R_i)$), the condition (4.2) reduces to $F(R_i|s_{-i}) > F(R_{-i}|s_{-i})$, which never holds for $\tilde{K}L(R_i||R_{-i}) = \tilde{K}L(R_{-i}||R_i)$ by (4.1). Hence, whenever the costs of mistakes do not differ across the two regimes, $F(R_i|s_i)$ decreases $\forall i$ as λ rises, which simply reflects decreased informativeness of signals due to increasing costs of learning.

Intuitively, consider the effects of the emergence of covid mutations (e.g. the omicron variant) following vaccination breakthroughs on the covid-induced level of uncertainty. Although the mutations raised new questions about the nature of covid and posed novel

¹²Whenever $\tilde{K}L(R_i||R_{-i}) > \tilde{K}L(R_{-i}||R_i)$, $\left(1 - e^{-\frac{1}{\lambda}\tilde{K}L(R_i||R_{-i})}\right) > 1 - e^{-\frac{1}{\lambda}\tilde{K}L(R_{-i}||R_i)}$ but $e^{-\frac{1}{\lambda}\tilde{K}L(R_{-i}||R_i)} < 1$.

¹³Suppose not. Then, $F(R_i|s_i) < F(R_{-i}|s_i) \forall i$ and so the agent can improve his information strategy costlessly by reinterpreting s_i as s_{-i} .

challenges in the fight against it, they did not increase the covid-induced uncertainty very much, since the global community already had a solid understanding of the virus and vaccination breakthroughs had already mitigated the potential perils of mutations. Hence, the appearance of Omicron variant could be thought of as a local rise in λ . At the same time, one could think of the stakes of mistakes to have been roughly the same in the two potential regimes - underestimating the Omicron variant in the event that it were severe could have incurred heavy economic costs, but so could overestimating its severity and imposing lockdowns on already stumbling economy. The responses of $F(R_{-i}|s_{-i})$, $i = 1, 2$, to changes in λ under the assumption of $\tilde{K}L(R_i||R_{-i}) = \tilde{K}L(R_{-i}||R_i)$ could thus be interpreted as the adjustments of the market reactions to good and bad news relative to the pre-Omicron state.

Whenever $\tilde{K}L(R_i||R_{-i}) \neq \tilde{K}L(R_{-i}||R_i)$, however, (4.2) might hold and $\frac{\partial}{\partial \lambda}(F(R_i|s_i))$ could be positive. To see the intuition behind this, note that $\tilde{K}L(R_i||R_{-i})$ is the cost of false negative about R_i , while $F(R_i, s_{-i})$ is the ex-ante probability of making such a mistake. $\tilde{K}L(R_i||R_{-i})F(R_i, s_{-i})$ is thus the ex-ante expected cost of false negatives about R_i , while $\tilde{K}L(R_{-i}||R_i)F(R_{-i}, s_{-i})$ is the ex-ante expected amount of costs avoided as a result of correctly identifying R_{-i} . Condition (4.2) then states that whenever the expected costs of missing regime R_i are higher than the expected benefits of correctly identifying R_{-i} , the optimal information strategy dictates that $F(R_i|s_i)$ increase. Whenever that is the case, $\frac{\partial}{\partial \lambda}(F(R_{-i}|s_{-i}))$ is always negative (otherwise $\frac{\partial}{\partial \lambda}(F(R_i|s_i)) > 0 \forall i$, which cannot hold if information strategy is optimal). Hence, an increase in $F(R_i|s_i)$ in response to a rise in λ always happens at the cost of decreased $F(R_{-i}|s_{-i})$, which in turn jointly reflects the trade-off between $F(R_i|s_i)$ and $F(R_{-i}|s_{-i})$ and the importance of avoiding mistakes in regime R_i . In words, whenever the probability-weighted cost of missing a disaster regime is higher than the probability-weighted cost of the disaster false positive (corresponding to the case of $i = 2$), the optimal information strategy dictates that the effort be shifted in response to a local increase in λ towards making information strategy more precise at identifying disaster regimes whenever present.

Part v) of Theorem 4.1 describes how regime beliefs translate into the beliefs about consumption growth outcomes. The result follows by Bayesian updating and the fact that s_λ is a signal about R only and conditional on knowing R , it provides no information about x_t . It states that the posterior probability of x_t^j after observing signal realization s_i is a convex combination of the probabilities of x_t^j induced by the two regimes, where the weights are given by the posterior regime probabilities corresponding

to s_i . Hence, all the insights about $F(R_k|s_i)$, $i, k = 1, 2$, are automatically transferred to $P(x_t^j|s_i)$ and through them, to equilibrium prices. This is illustrated in the section that follows.

5 The Equilibrium

Theorem 5.1. *Let $P_{Z,0}$ denote the time-zero observed price of an asset with pay-off function $Z(x_t)$ following a signal realization s_i and when the cost of information is λ . Let $P_{Z,0}^i$ denote the price of the asset in regime R_i if it were fully observable, and $F(R_i|s_i)$ - subjective posterior probability of R_i after observing s_i under optimal information strategy. Then,*

$$P_{Z,0} = F(R_i|s_i) P_{Z,0}^i + F(R_{-i}|s_i) P_{Z,0}^{-i}.$$

Proof. See the Appendix. □

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Appendix

Portfolio Selection Problem With Arbitrary Beliefs

$c(x_t) = \left((c(x_t^j))_{j=1}^{t+1} \right)_{t=0}^T$ and $q(x_t^j)$ also without j and $c(x_t^j)$ denote, respectively, the state-price and Arrow-Debreu security corresponding to x_t^j . let q denote state prices and c ad securities. Let $P(x_t)$ be arbitrary beliefs. a rep investor want to for fixed state prices

$$\max_{c(x_t)} \sum_{t=0}^T \beta^t \sum_{j=1}^{t+1} \log(c(x_t^j)) P(x_t^j)$$

subject to

$$\sum_{t=0}^T \sum_{j=1}^{t+1} q(x_t^j) c(x_t^j) \leq \sum_{t=0}^T \sum_{j=1}^{t+1} q(x_t^j) \omega(x_t^j).$$

The Lagrangian of the optimization problem is given by

$$\begin{aligned} \mathcal{L}(c(x_t), \mu) &= \sum_{t=0}^T \beta^t \sum_{j=1}^{t+1} \log(c(x_t^j)) P(x_t^j) + \\ &+ \mu \left(\sum_{t=0}^T \sum_{j=1}^{t+1} q(x_t^j) \omega(x_t^j) - \sum_{t=0}^T \sum_{j=1}^{t+1} q(x_t^j) c(x_t^j) \right) \end{aligned}$$

and the First Order Condition w.r.t $c(x_t^j)$ gives

$$\begin{aligned} &= \mu q(x_t^j) \\ c(x_t^j) &= \frac{1}{\mu} \frac{\beta^t P(x_t^j)}{q(x_t^j)}. \end{aligned}$$

If we set $q(x_0)$ as numeraire, we get that

$$c(x_0) = \frac{1}{\mu}$$

and so

$$c(x_t^j) = c(x_0) \frac{\beta^t P(x_t^j)}{q(x_t^j)}. \quad (6.1)$$

Finally, the cost of $c(x_t)$ given q is given by

$$\begin{aligned} \sum_{t=0}^T \sum_{j=1}^{t+1} q(x_t^j) c(x_t^j) &= \sum_{t=0}^T \sum_{j=1}^{t+1} q(x_t^j) c(x_0) \frac{\beta^t P(x_t^j)}{q(x_t^j)} = \\ &= c(x_0) \sum_{t=0}^T \beta^t \end{aligned}$$

and by the budget constraint,

$$c(x_0) = \left(\sum_{t=0}^T \sum_{j=1}^{t+1} q(x_t^j) \omega(x_t^j) \right) / \sum_{t=0}^T \beta^t. \quad (6.2)$$

Since the objective function is concave in choice variables, while the constraint - linear, $c(x_t)$ defined by (6.2) and (6.1) is utility-maximizing.

Solution to the Signal Selection Problem

Summary of the Solution

Henceforth, let R_1 denote the Good regime, R_2 - the Disaster Regime, s_i - a signal realization that suggests that the true regime is R_i , and set $\mathcal{P}_G \equiv \mathcal{P}_1$, $\mathcal{P}_D \equiv \mathcal{P}_2$. The solution to the signal selection problem can be summarized as follows: I first describe the RI framework of Matejka and McKay (2015) and establish its mapping with the signal selection problem of the representative investor, which leads to the following non-reduced-form expression describing the posterior beliefs about R after observing s_i :

$$F(R|s_i) = G(R) \frac{e^{V(\mathcal{P}_i, R)/\lambda}}{\sum_{k=1}^2 p_{s_k} e^{V(\mathcal{P}_k, R)/\lambda}}, \quad \forall i, R, \quad (6.3)$$

where p_k is the unconditional probability of observing s_k . $\frac{e^{V(\mathcal{P}_i, R)/\lambda}}{\sum_{k=1}^2 p_{s_k} e^{V(\mathcal{P}_k, R)/\lambda}} \equiv w(i, R)$ is the posterior weight corresponding to regime R and signal realization i , i.e. $w(i, R)$ is the number by which one needs to multiply the prior probability of R after observing s_i to get $F(R|s_i)$ under optimal information strategy. In this form, however, the posterior probabilities are not very useful, since $V(\mathcal{P}_k, R)$ depends on state prices (see expression (6.7)) and p_k is not in reduced form (i.e. (6.3) establishes the relationship between $F(R|s_i)$ and p_k under optimal information strategy, but does not pin down an

expression for p_k). In the next step (Lemma 6.1), I show that

$$w(i, R) = \frac{e^{-\frac{1}{\lambda} \tilde{K}L(R||R_i)}}{\sum_{k=1}^2 p_{s_k} e^{-\frac{1}{\lambda} \tilde{K}L(R||R_k)}},$$

i.e. the state prices in the numerator and the denominator of the RHS of (6.3) cancel each other and $V(\mathcal{P}_k, R)$ gets replaced by $-\tilde{K}L(R||R_k)$, which is a function of exogenous parameters only. Lastly, I obtain p_{s_k} in reduced form (Proposition 6.1) and show that

$$p_{s_\lambda} = G(R_1) \frac{1}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}} - G(R_2) \frac{e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}$$

and

$$F(R_i|s_i) = \frac{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_{-i}||R_i)}}{1 - e^{-\frac{1}{\lambda} (\tilde{K}L(R_2||R_1) + \tilde{K}L(R_1||R_2))}} \quad \forall i, R,$$

where p_{s_λ} is p_{s_1} under optimal information strategy. The last result is proved under Proposition 6.2.

Rational Inattention Discrete Choice Framework of Matejka and McKay (2015)

In the discrete choice framework of Matejka and McKay (2015) (Section IA), the Decision-Maker (DM) chooses among N alternative actions with pay-off vector $\mathbf{v}$ (also called the state of nature), where v_i - the i -th entry of $\mathbf{v}$, is the pay-off of i -th action. The DM does not observe $\mathbf{v}$, but has prior beliefs $G(\mathbf{v})$ about it and wants to select a costly signal $\mathbf{S} \in \mathbb{R}^N$ proxying $\mathbf{v}$ and such that if he were to choose the action with the highest posterior pay-off after observing a signal realization $\mathbf{s}$ of S , no other signal function would generate higher ex-ante expected pay-off net of the cost of information.

Mathematically, given the information strategy F (equivalently, the joint distribution of $\mathbf{S}$ and $\mathbf{v}$) and a signal realization $\mathbf{s}$, the DM chooses action i such that

$$i = \arg \max_j E_{F(\mathbf{v}|\mathbf{s})}(v_j)$$

and wants to select information strategy F that solves

$$\max_F \left\{ \sum_{\mathbf{v}} \sum_{\mathbf{s}} V(F(\cdot|\mathbf{s})) F(\mathbf{s}|\mathbf{v}) G(\mathbf{v}) - c(F) \right\}$$

subject to

$$\sum_{\mathbf{s}} F(\mathbf{s}, \mathbf{v}) = G(\mathbf{v}) \quad \forall \mathbf{v},$$

where $V(F(\cdot|\mathbf{s})) \equiv \max_j E_{F(\mathbf{v}|\mathbf{s})}(v_j) = E_{F(\mathbf{v}|\mathbf{s})}(v_i)$ and $c(F)$ is defined as in (3.2). Matejka and McKay (2015) show that under optimal strategy there must be one-to-one correspondence between actions and signals (LEMMA 1), since exerting effort to differentiate among signals that lead to the same action incurs costs but does not affect the pay-off. Using this result, letting s_i denote a signal realization that leads to action i and setting $\mathcal{P}_i(\mathbf{v}) \equiv F(s_i|\mathbf{v})$, Matejka and McKay (2015) show in Corollary 1 of their paper that the information choice problem can equivalently be formulated as

$$\max_{\mathcal{P}=(\mathcal{P}_i(\mathbf{v}))_{i=1}^N} \left\{ \sum_{i=1}^N \sum_{\mathbf{v}} v_i \mathcal{P}_i(\mathbf{v}) G(\mathbf{v}) - c(G, \mathcal{P}) \right\} \quad (6.4)$$

subject to

$$\begin{aligned} \forall i: \quad \mathcal{P}_i(\mathbf{v}) &\geq 0 \quad \forall \mathbf{v} \in \mathcal{S}_G \\ \sum_i \mathcal{P}_i(\mathbf{v}) &= 1 \quad \forall \mathbf{v} \in \mathcal{S}_G, \end{aligned}$$

where $\mathcal{S}_G$ is the support of the prior beliefs and $c(G, \mathcal{P})$ is $c(F)$ reparametrized in terms of $\mathcal{P}_i(\mathbf{v})$. By Theorem 1 in Section IIA of Matejka and McKay (2015), the conditional probabilities of the optimal information strategy satisfy

$$\mathcal{P}_i(\mathbf{v}) = \frac{\mathcal{P}_i^0 e^{v_i/\lambda}}{\sum_{j=1}^N \mathcal{P}_j^0 e^{v_j/\lambda}} \quad \forall i, \mathbf{v}, \quad (6.5)$$

where $\mathcal{P}_i^0 = \sum_{\mathbf{v}} \mathcal{P}_i(\mathbf{v}) G(\mathbf{v})$ is the unconditional probability of observing s_i . Note that the expressions above do not completely describe the information strategy, since conditional probabilities $(\mathcal{P}_i(\mathbf{v}))_i$ depend on unconditional probabilities $(\mathcal{P}_i^0)_i$, which in turn depend on $(\mathcal{P}_i(\mathbf{v}))_i$. The full solution of the optimal probabilities requires additional manipulations (see COROLLARY 2 and ensuing discussion in Matejka and McKay (2015)) and cannot in general be obtained in closed-form. The solution to the signal selection problem of the representative investor, however, is a special case of this framework in which closed-form solutions can be obtained.

Mapping Between Signal Selection Problem and RI Framework of Matejka and McKay (2015)

Note that intuitively $V(\mathcal{P}_s, R)$ is the pay-off of a particular action (choosing $\mathcal{P}_s$) in a particular state of nature (or regime R of the economy). This is the observation that motivates the formal mapping between the RI framework of Matejka and McKay (2015) and the signal selection problem of the representative investor.

Let $N = 2$ in the RI framework of Matejka and McKay (2015) and let

$$\mathbf{v}_1 = \begin{bmatrix} V(\mathcal{P}_G, G) \\ V(\mathcal{P}_D, G) \end{bmatrix}$$

and

$$\mathbf{v}_2 = \begin{bmatrix} V(\mathcal{P}_G, D) \\ V(\mathcal{P}_D, D) \end{bmatrix}.$$

Since $V(\mathcal{P}_G, G)$ and $V(\mathcal{P}_D, G)$ are Good-regime pay-offs of choosing $\mathcal{P}_G$ and $\mathcal{P}_D$, respectively, and $V(\mathcal{P}_G, D)$ and $V(\mathcal{P}_D, D)$ are the Disaster-regime pay-offs of choosing $\mathcal{P}_G$ and $\mathcal{P}_D$, respectively, action i can be interpreted as choosing portfolio i (with $\mathcal{P}_G$ denoting portfolio 1 and $\mathcal{P}_D$ denoting portfolio 2), while the state of nature can be interpreted as the regime R the economy is in. To see that such interpretation is mathematically valid, note that the entries within each $\mathbf{v}$ vary only through the first argument of $V(\mathcal{P}_s, R)$, and $\mathbf{v}_1 = \mathbf{v}_2$ iff the two regimes are the same¹⁴. Hence, there is a one-to-one correspondence between $\mathbf{v}$ and R and so any signal about $\mathbf{v}$ can be interpreted as a signal about R . Moreover, since s_i is defined in Matejka and McKay (2015) as a signal realization that leads to action i , s_1 can be interpreted as s_G , a signal realization that suggests $R = G$, and s_2 can be interpreted as s_D , a signal realization that suggests $R = D$. Such interpretation is further justified by the fact that $\mathcal{P}_G$ has higher expected pay-off in the Good regime, while $\mathcal{P}_D$ has higher expected pay-off in the Disaster regime.

Imposing this additional structure and letting $\mathcal{P}_s(R) \equiv F(s|R)$, (6.4) can immediately be rewritten as

$$\max_{\mathcal{P}=(\mathcal{P}_s(R))_{i=1}^N} \left\{ \sum_{s=s_G, s_D} \sum_{R=G, D} V(\mathcal{P}_s, R) \mathcal{P}_s(R) G(R) - c(G, \mathcal{P}) \right\}$$

¹⁴If part is obvious. For the only if part, note that since the portfolio selection problem has a unique solution, $V(\mathcal{P}_G, G) > V(\mathcal{P}_D, G)$ and $V(\mathcal{P}_G, D) < V(\mathcal{P}_D, D)$. Hence, $\mathbf{v}_1 \neq \mathbf{v}_2$.

subject to

$$\forall s: \quad \mathcal{P}_s(R) \geq 0 \quad R = G, D$$

$$\sum_s \mathcal{P}_s(R) = 1 \quad R = G, D,$$

Let p_s denote the prior probability of observing a signal realization s . Since the additional structure does not impose any restrictions on the RI framework of Matejka and McKay (2015) and only adds interpretative insight, it is immediate by (6.5) that

$$F(s_i|R) = \frac{p_{s_i} e^{V(\mathcal{P}_i, R)/\lambda}}{\sum_s p_s e^{V(\mathcal{P}_s, R)/\lambda}}, \quad \forall i, R.$$

Finally, by Bayes Law,

$$F(R|s_i) = G(R) \frac{e^{V(\mathcal{P}_i, R)/\lambda}}{\sum_s p_s e^{V(\mathcal{P}_s, R)/\lambda}}, \quad \forall i, R. \quad (6.6)$$

Reduced-Form Expressions for Conditional Signal Probabilities

Lemma 6.1. *Let $V(\mathcal{P}_i, R)$ denote the expected utility of portfolio i in Regime R and let $\tilde{K}L(R||R_k) \equiv \sum_{t=1}^T \beta^t KL(P(x_t|R)||P(x_t|R_k))$. Let s_k denote a signal realization that suggests that the true regime is R_k and let p_{s_k} be the unconditional probability of observing s_k , $k = 1, 2$. Then,*

$$\frac{e^{V(\mathcal{P}_i, R)/\lambda}}{\sum_{k=1}^2 p_{s_k} e^{V(\mathcal{P}_k, R)/\lambda}} = \frac{e^{-\frac{1}{\lambda} \tilde{K}L(R||R_i)}}{\sum_{k=1}^2 p_{s_k} e^{-\frac{1}{\lambda} \tilde{K}L(R||R_k)}} \quad \forall i, R.$$

Proof. The proof is structured as follows: First, I derive an exact expression for $V(\mathcal{P}_i, R)$ and show that it can be expressed as a sum of the terms that depend on i and the terms that do not depend on i . Next, I show that this allows us to simplify $\frac{e^{V(\mathcal{P}_i, R)/\lambda}}{\sum_{k=1}^2 p_{s_k} e^{V(\mathcal{P}_k, R)/\lambda}}$ by canceling terms in the numerator and the denominator and finally, I carry out additional manipulations to express whatever remains in terms of relative entropies.

Take arbitrary i and R and start by noting that $c_i(x_t) \equiv \mathcal{P}_i$, i.e. $c_i(x_t)$ given by (6.1) and (6.2) is the optimal portfolio under beliefs $P(x_t^j) = P(x_t^j|R_i)$. Thus,

$$V(\mathcal{P}_i, R) = \sum_{t=0}^T \beta^t \sum_{j=1}^{t+1} \log(c_i(x_t^j)) P(x_t^j|R) =$$

$$\begin{aligned}
&= \log(c_i(x_0)) + \sum_{t=1}^T \beta^t \sum_{j=1}^{t+1} \log(c_i(x_t^j)) P(x_t^j|R) = \\
&= \log(c_i(x_0)) + \sum_{t=1}^T \beta^t \sum_{j=1}^{t+1} \log\left(c_i(x_0) \frac{\beta^t P(x_t^j|R_i)}{q(x_t^j)}\right) P(x_t^j|R) = \\
&= \log(c(x_0)) + \log \prod_{t=1}^T \prod_{j=1}^{t+1} \left(c(x_0) \frac{\beta^t P(x_t^j|R_i)}{q(x_t^j)}\right)^{\beta^t P(x_t^j|R)}, \tag{6.7}
\end{aligned}$$

where the first equality follows by definition, the second - by trivial algebra, the third - by (6.1), and the last - by the properties of the logarithmic function. $c_i(x_0)$ gets replaced by $c(x_0)$ on the last line, since by (6.2), $c_i(x_0)$ does not depend on the underlying probability distribution and hence does not vary across i (or signal realizations) for fixed $q(x_t)$. Next, note that

$$\begin{aligned}
&\prod_{t=1}^T \prod_{j=1}^{t+1} \left(c(x_0) \frac{\beta^t P(x_t^j|R_i)}{q(x_t^j)}\right)^{\beta^t P(x_t^j|R)} = \\
&= \prod_{t=1}^T (\beta^t c(x_0))^{\beta^t} \prod_{j=1}^{t+1} \left(\frac{1}{q(x_t^j)}\right)^{\beta^t P(x_t^j|R)} \prod_{j=1}^{t+1} (P(x_t^j|R_i))^{\beta^t P(x_t^j|R)} \equiv \\
&\equiv A \prod_{t=1}^T \prod_{j=1}^{t+1} (P(x_t^j|R_i))^{\beta^t P(x_t^j|R)}, \tag{6.8}
\end{aligned}$$

where $A \equiv \prod_{t=1}^T (\beta^t c(x_0))^{\beta^t} \prod_{j=1}^{t+1} \left(\frac{1}{q(x_t^j)}\right)^{\beta^t P(x_t^j|R)}$ collects terms that do not depend on i . Putting (6.7) and (6.8) together, we further get

$$\begin{aligned}
V(\mathcal{P}_i, R) &= \log(c(x_0)) + \log A \prod_{t=1}^T \prod_{j=1}^{t+1} (P(x_t^j|R_i))^{\beta^t P(x_t^j|R)} = \\
&= \log(c(x_0)) + \log A + \log \prod_{t=1}^T \prod_{j=1}^{t+1} (P(x_t^j|R_i))^{\beta^t P(x_t^j|R)} = \\
&= \log(c(x_0)) + \log A + \sum_{t=1}^T \beta^t \sum_{j=1}^{t+1} P(x_t^j|R) \log(P(x_t^j|R_i)) =
\end{aligned}$$

$$= \log(c(x_0)) + \log A + \sum_{t=1}^T \beta^t E_{R,t}[\log(P(x_t|R_i))], \quad (6.9)$$

where $E_{R,t}[\cdot]$ denotes an expectation object over time- t outcomes induced by beliefs $P(x_t|R)$. Inserting (6.11) back into $\frac{e^{V(\mathcal{P}_i,R)/\lambda}}{\sum_{k=1}^2 p_{s_k} e^{V(\mathcal{P}_k,R)/\lambda}}$ and noting that $\log(c(x_0)) + \log A$ does not depend on a signal realization s_k , we get by straightforward algebra that

$$\begin{aligned} \frac{e^{V(\mathcal{P}_i,R)/\lambda}}{\sum_{k=1}^2 p_{s_k} e^{V(\mathcal{P}_k,R)/\lambda}} &= \frac{e^{\left(\log(c(x_0)) + \log A + \sum_{t=1}^T \beta^t E_{R,t}[\log(P(x_t|R_i))]\right)/\lambda}}{\sum_{k=1}^2 p_{s_k} e^{\left(\log(c(x_0)) + \log A + \sum_{t=1}^T \beta^t E_{R,t}[\log(P(x_t|R_k))]\right)/\lambda}} = \\ &= \frac{e^{\frac{1}{\lambda}(\log(c(x_0)) + \log A) + \frac{1}{\lambda} \sum_{t=1}^T \beta^t E_{R,t}[\log(P(x_t|R_i))]}{\sum_{k=1}^2 p_{s_k} e^{\frac{1}{\lambda}(\log(c(x_0)) + \log A) + \frac{1}{\lambda} \sum_{t=1}^T \beta^t E_{R,t}[\log(P(x_t|R_k))]} = \\ &= \frac{e^{\frac{1}{\lambda}(\log(c(x_0)) + \log A)} e^{\frac{1}{\lambda} \sum_{t=1}^T \beta^t E_{R,t}[\log(P(x_t|R_i))]}{e^{\frac{1}{\lambda}(\log(c(x_0)) + \log A)} \sum_{k=1}^2 p_{s_k} e^{\frac{1}{\lambda} \sum_{t=1}^T \beta^t E_{R,t}[\log(P(x_t|R_k))]} = \\ &= \frac{e^{\frac{1}{\lambda} \sum_{t=1}^T \beta^t E_{R,t}[\log(P(x_t|R_i))]}{\sum_{k=1}^2 p_{s_k} e^{\frac{1}{\lambda} \sum_{t=1}^T \beta^t E_{R,t}[\log(P(x_t|R_k))]} \quad (6.10) \end{aligned}$$

Further, dividing both the numerator and the denominator of (6.10) by $e^{\frac{1}{\lambda} \sum_{t=1}^T \beta^t E_{R,t}[\log(P(x_t|R))]}$, we get

$$\begin{aligned} \frac{e^{V(\mathcal{P}_i,R)/\lambda}}{\sum_{k=1}^2 p_{s_k} e^{V(\mathcal{P}_k,R)/\lambda}} &= \frac{e^{\frac{1}{\lambda} \sum_{t=1}^T \beta^t E_{R,t}[\log(P(x_t|R_i))] - \frac{1}{\lambda} \sum_{t=1}^T \beta^t E_{R,t}[\log(P(x_t|R))]}{\sum_{k=1}^2 p_{s_k} e^{\frac{1}{\lambda} \sum_{t=1}^T \beta^t E_{R,t}[\log(P(x_t|R_k))] - \frac{1}{\lambda} \sum_{t=1}^T \beta^t E_{R,t}[\log(P(x_t|R))]} = \\ &= \frac{e^{\frac{1}{\lambda} \sum_{t=1}^T \beta^t (E_{R,t}[\log(P(x_t|R_i))] - E_{R,t}[\log(P(x_t|R))])}}{\sum_{k=1}^2 p_{s_k} e^{\frac{1}{\lambda} \sum_{t=1}^T \beta^t (E_{R,t}[\log(P(x_t|R_k))] - E_{R,t}[\log(P(x_t|R))])}} = \\ &= \frac{e^{-\frac{1}{\lambda} \sum_{t=1}^T \beta^t KL(P(x_t|R) || P(x_t|R_i))}}{\sum_{k=1}^2 p_{s_k} e^{-\frac{1}{\lambda} \sum_{t=1}^T \beta^t KL(P(x_t|R) || P(x_t|R_k))}}, \end{aligned}$$

where $KL(P(x_t|R) || P(x_t|R_i)) = E_{R,t} \left[\frac{\log(P(x_t|R))}{\log(P(x_t|R_i))} \right]$ is a Kullback-Leibler divergence from $P(x_t|R)$ to $P(x_t|R_i)$. Finally, setting

$$\tilde{KL}(R || R_k) \equiv \sum_{t=1}^T \beta^t KL(P(x_t|R) || P(x_t|R_k)), \quad k = 1, 2,$$

it follows that

$$\frac{e^{V(\mathcal{P}_i, R)/\lambda}}{\sum_{k=1}^2 p_{s_k} e^{V(\mathcal{P}_k, R)/\lambda}} = \frac{e^{-\frac{1}{\lambda} \tilde{K}L(R||R_i)}}{\sum_{k=1}^2 p_{s_k} e^{-\frac{1}{\lambda} \tilde{K}L(R||R_k)}}.$$

□

Proposition 6.1. *Let p_{s_λ} denote the probability of observing s_1 under the optimal information strategy and let p_0 denote the prior perceived probability of R_1 . Then,*

$$p_{s_\lambda} = p_0 \frac{1}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}} - (1 - p_0) \frac{e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}.$$

Proof. By (6.6) and Lemma 6.1,

$$p_0 \frac{1}{p_{s_\lambda} + (1 - p_{s_\lambda}) e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}} + (1 - p_0) \frac{e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}}{p_{s_\lambda} e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} + (1 - p_{s_\lambda})} = 1$$

and

$$p_0 \frac{e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}{p_{s_\lambda} + (1 - p_{s_\lambda}) e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}} + (1 - p_0) \frac{1}{p_{s_\lambda} e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} + (1 - p_{s_\lambda})} = 1.$$

Multiplying both sides of both equations above, we get

$$(1 - p_0) \frac{p_{s_\lambda} + (1 - p_{s_\lambda}) e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}{p_{s_\lambda} e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} + (1 - p_{s_\lambda})} = \frac{p_{s_\lambda} + (1 - p_{s_\lambda}) e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} - p_0}{e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}}$$

and

$$(1 - p_0) \frac{p_{s_\lambda} + (1 - p_{s_\lambda}) e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}{p_{s_\lambda} e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} + (1 - p_{s_\lambda})} = p_{s_\lambda} + (1 - p_{s_\lambda}) e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} - p_0 e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}.$$

Since the LHS is the same in both equations, we have

$$\frac{p_{s_\lambda} + (1 - p_{s_\lambda}) e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} - p_0}{e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}} = p_{s_\lambda} + (1 - p_{s_\lambda}) e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} - p_0 e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}$$

and so

$$\begin{aligned} p_{s_\lambda} + (1 - p_{s_\lambda}) e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} - p_0 &= p_{s_\lambda} e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} + \\ &+ (1 - p_{s_\lambda}) e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} - p_0 e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}. \end{aligned}$$

Moving p_{s_λ} terms to the LHS, we further get

$$\begin{aligned} & p_{s_\lambda} \left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} + e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} \right) = \\ & = -e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} + p_0 + e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} - p_0 e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}, \end{aligned}$$

which implies by straightforward algebra that

$$\begin{aligned} & p_{s_\lambda} \left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} \right) \left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} \right) = \\ & = p_0 \left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} \right) - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} \left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} \right). \quad (6.11) \end{aligned}$$

Further, note that

$$\begin{aligned} & e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} \left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} \right) = (p_0 + 1 - p_0) e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} \left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} \right) = \\ & = p_0 e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} \left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} \right) + (1 - p_0) e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} \left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} \right) \end{aligned}$$

and so, inserting this back into (6.11), we get

$$\begin{aligned} & p_{s_\lambda} \left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} \right) \left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} \right) = \\ & = p_0 \left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} \right) - (1 - p_0) e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} \left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} \right). \end{aligned}$$

Finally, by basic algebra,

$$\begin{aligned} p_{s_\lambda} &= \frac{p_0 \left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} \right) - (1 - p_0) e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} \left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} \right)}{\left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} \right) \left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} \right)} = \\ &= p_0 \frac{1}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}} - (1 - p_0) \frac{e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}. \quad (6.12) \end{aligned}$$

□

Corollary 6.1. *Under optimal information strategy,*

$$\lim_{\lambda \rightarrow 0} p_{s_\lambda} = p_0.$$

Proof. Since $\tilde{K}L(R_2||R_1) > 0$,

$$\lim_{\lambda \rightarrow 0} e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} = e^{-\tilde{K}L(R_2||R_1) \lim_{\lambda \rightarrow 0} \frac{1}{\lambda}} = e^{-\infty} = 0$$

and so

$$\lim_{\lambda \rightarrow 0} p_{s_\lambda} = p_0 \frac{1}{1 - \lim_{\lambda \rightarrow 0} e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}} - (1 - p_0) \frac{\lim_{\lambda \rightarrow 0} e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}{1 - \lim_{\lambda \rightarrow 0} e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}} = p_0.$$

□

With unconditional optimal signal probabilities obtained in closed-form, we are now ready to derive reduced-form posterior probabilities of R given s . This is done in the proposition that follows.

Proposition 6.2. *Under optimal information strategy,*

$$F(R_i|s_i) = \frac{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_{-i}||R_i)}}{1 - e^{-\frac{1}{\lambda} (\tilde{K}L(R_2||R_1) + \tilde{K}L(R_1||R_2))}} \quad i = 1, 2.$$

Proof. The proof is based on inserting the results of Proposition 6.1 into the result of Lemma 6.1. Start by noting that by Proposition 6.1,

$$\begin{aligned} 1 - p_{s_\lambda} &= \frac{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}} - \\ &- p_0 \frac{1}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}} + (1 - p_0) \frac{e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}} = \\ &= \frac{1 - p_0 e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}} - p_0 \frac{1}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}} \end{aligned}$$

and so

$$(1 - p_{s_\lambda}) e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} = \frac{1 - p_0 e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}} e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} - p_0 \frac{e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}}$$

and

$$p_{s_\lambda} + (1 - p_{s_\lambda}) e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} = p_0 \frac{1}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}} - (1 - p_0) \frac{e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}} +$$

$$\begin{aligned}
& + \frac{1 - p_0 e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}} e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} - p_0 \frac{e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}} = \\
& = p_0 \frac{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}} + p_0 e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} = \\
& = p_0 \frac{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} + \left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}\right) e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}} = \\
& = p_0 \frac{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}}, \tag{6.13}
\end{aligned}$$

where the second equality follows by adding up terms with the common denominator and the rest is trivial algebra. Note that (6.13) is the denominator of the posterior weight $w(i, R)$ corresponding to $R = R_1$ and arbitrary i .¹⁵ Hence, by Lemma 6.1 and (6.13),

$$w(i, R_1) = \frac{e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_i)}}{\sum_{k=1}^2 p_{s_k} e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_k)}} = \frac{e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_i)}}{p_0} \frac{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}$$

and since

$$\tilde{K}L(R_1||R_1) = \sum_{t=1}^T \beta^t KL(P(x_t|R_1)||P(x_t|R_1)) = 0,$$

$$w(1, R_1) = \frac{1}{p_0} \frac{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}$$

and

$$F(R_1|s_1) = p_0 w(1, R_1) = \frac{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}.$$

Likewise,

$$w(2, R_1) = e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)} \frac{1}{p_0} \frac{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)}}{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_2||R_1)} e^{-\frac{1}{\lambda} \tilde{K}L(R_1||R_2)}}$$

and we get finally that

$$F(R_2|s_2) = 1 - F(R_1|s_2) = 1 - p_0 w(2, R_1) =$$

¹⁵Recall that the posterior weight $w(i, R)$ is the number by which one needs to scale the prior about R after observing s_i to get the posterior probability of R under optimal information strategy.

$$\begin{aligned}
&= \frac{1 - e^{-\frac{1}{\lambda}\tilde{K}L(R_2||R_1)}e^{-\frac{1}{\lambda}\tilde{K}L(R_1||R_2)}}{1 - e^{-\frac{1}{\lambda}\tilde{K}L(R_2||R_1)}e^{-\frac{1}{\lambda}\tilde{K}L(R_1||R_2)}} - \frac{e^{-\frac{1}{\lambda}\tilde{K}L(R_1||R_2)} - e^{-\frac{1}{\lambda}\tilde{K}L(R_2||R_1)}e^{-\frac{1}{\lambda}\tilde{K}L(R_1||R_2)}}{1 - e^{-\frac{1}{\lambda}\tilde{K}L(R_2||R_1)}e^{-\frac{1}{\lambda}\tilde{K}L(R_1||R_2)}} = \\
&= \frac{1 - e^{-\frac{1}{\lambda}\tilde{K}L(R_1||R_2)}}{1 - e^{-\frac{1}{\lambda}\tilde{K}L(R_2||R_1)}e^{-\frac{1}{\lambda}\tilde{K}L(R_1||R_2)}}.
\end{aligned}$$

□

Corollary 6.2. *Under optimal information strategy,*

$$\lim_{\lambda \rightarrow 0} F(R_i|s_i) = \lim_{\lambda \rightarrow 0} F(s_i|R_i) = 1, \quad i = 1, 2.$$

Proof. Note first that since $\forall a \in \mathbb{R}^+$,

$$\lim_{\lambda \rightarrow 0} e^{-\frac{1}{\lambda}a} = e^{-a \times \infty} = 0,$$

we have

$$\lim_{\lambda \rightarrow 0} F(R_i|s_i) = \frac{1 - \lim_{\lambda \rightarrow 0} e^{-\frac{1}{\lambda}\tilde{K}L(R_{-i}||R_i)}}{1 - \lim_{\lambda \rightarrow 0} e^{-\frac{1}{\lambda}(\tilde{K}L(R_2||R_1) + \tilde{K}L(R_1||R_2))}} = 1.$$

Next, let $p_{s_i, \lambda}$ denote the unconditional probability of observing a signal realization s_i and note by Corollary 6.1 that

$$\lim_{\lambda \rightarrow 0} p_{s_i, \lambda} = G(R_i), \quad i = 1, 2.$$

It then follows by Bayesian updating and limit laws that

$$\begin{aligned}
\lim_{\lambda \rightarrow 0} F(s_i|R_i) &= \lim_{\lambda \rightarrow 0} \frac{F(R_i|s_i)p_{s_i, \lambda}}{G(R_i)} = \frac{\lim_{\lambda \rightarrow 0} F(R_i|s_i) \lim_{\lambda \rightarrow 0} p_{s_i, \lambda}}{G(R_i)} = \\
&= \frac{G(R_i)}{G(R_i)} = 1.
\end{aligned}$$

The claim of the Corollary is then immediate. □

Corollary 6.3. *Under optimal information strategy,*

$$\frac{\partial}{\partial \lambda} (F(R_i|s_i)) > 0$$

iff

$$\tilde{K}L(R_i||R_{-i})F(R_i, s_{-i}) > \tilde{K}L(R_{-i}||R_i)F(R_{-i}, s_{-i}).$$

Proof. Note first that by straightforward differentiation,

$$\begin{aligned} & \frac{\partial}{\partial \lambda} \left(\frac{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_{-i}||R_i)}}{1 - e^{-\frac{1}{\lambda} (\tilde{K}L(R_2||R_1) + \tilde{K}L(R_1||R_2))}} \right) = \\ & = \frac{-\tilde{K}L(R_{-i}||R_i) \lambda^{-2} e^{-\frac{1}{\lambda} \tilde{K}L(R_{-i}||R_i)} \left(1 - e^{-\frac{1}{\lambda} (\tilde{K}L(R_{-i}||R_i) + \tilde{K}L(R_i||R_{-i}))} \right)}{\left(1 - e^{-\frac{1}{\lambda} (\tilde{K}L(R_{-i}||R_i) + \tilde{K}L(R_i||R_{-i}))} \right)^2} + \\ & + \frac{\left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_{-i}||R_i)} \right) \left(\tilde{K}L(R_{-i}||R_i) + \tilde{K}L(R_i||R_{-i}) \right) \lambda^{-2} e^{-\frac{1}{\lambda} (\tilde{K}L(R_{-i}||R_i) + \tilde{K}L(R_i||R_{-i}))}}{\left(1 - e^{-\frac{1}{\lambda} (\tilde{K}L(R_{-i}||R_i) + \tilde{K}L(R_i||R_{-i}))} \right)^2}. \end{aligned}$$

Since the denominator is a squared quantity, $\frac{\partial F(R_i|s_i)}{\partial \lambda} > 0$ iff its numerator (*Num* henceforth) is positive, i.e. iff

$$\begin{aligned} Num = \lambda^{-2} & \left(-\tilde{K}L(R_{-i}||R_i) e^{-\frac{1}{\lambda} \tilde{K}L(R_{-i}||R_i)} + \tilde{K}L(R_{-i}||R_i) e^{-\frac{1}{\lambda} (\tilde{K}L(R_{-i}||R_i) + \tilde{K}L(R_i||R_{-i}))} + \right. \\ & \left. + \tilde{K}L(R_i||R_{-i}) \left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_{-i}||R_i)} \right) e^{-\frac{1}{\lambda} (\tilde{K}L(R_{-i}||R_i) + \tilde{K}L(R_i||R_{-i}))} \right) > 0. \end{aligned}$$

Dividing both sides by $\lambda^{-2} e^{-\frac{1}{\lambda} \tilde{K}L(R_{-i}||R_i)}$, we get that $Num > 0$ iff

$$\begin{aligned} & -\tilde{K}L(R_{-i}||R_i) \left(1 - e^{-\frac{1}{\lambda} (\tilde{K}L(R_i||R_{-i}))} \right) + \\ & + \tilde{K}L(R_i||R_{-i}) \left(1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_{-i}||R_i)} \right) e^{-\frac{1}{\lambda} (\tilde{K}L(R_i||R_{-i}))} > 0. \end{aligned}$$

Further dividing both sides by $1 - e^{-\frac{1}{\lambda} (\tilde{K}L(R_{-i}||R_i) + \tilde{K}L(R_i||R_{-i}))}$, we get that $Num > 0$ iff

$$\tilde{K}L(R_i||R_{-i}) (1 - F(R_{-i}|s_{-i})) - \tilde{K}L(R_{-i}||R_i) F(R_{-i}|s_{-i}) > 0,$$

which can be expressed using Bayes Law as

$$\tilde{K}L(R_i||R_{-i}) F(R_i, s_{-i}) > \tilde{K}L(R_{-i}||R_i) F(R_{-i}, s_{-i}).$$

□

Proposition 6.3. *The following are true under optimal information strategy:*

$$i) P(x_t^j | s_i) = P(x_t^j | R_i) \frac{1 - e^{-\frac{1}{\lambda} \tilde{K}L(R_{-i}||R_i)}}{1 - e^{-\frac{1}{\lambda} (\tilde{K}L(R_2||R_1) + \tilde{K}L(R_1||R_2))}} +$$

$$+P(x_t^j|R_{-i})\left(1-\frac{1-e^{-\frac{1}{\lambda}\tilde{K}L(R_{-i}|R_i)}}{1-e^{-\frac{1}{\lambda}(\tilde{K}L(R_2|R_1)+\tilde{K}L(R_1|R_2))}}\right), \quad i=1,2$$

$$ii) \lim_{\lambda \rightarrow 0} P(x_t^j|s_i) = P(x_t^j|R_i).$$

Proof. i) follows by the chain of equalities below:

$$\begin{aligned} P(x_t^j|s_i) &= \sum_{k=i,-i} P(x_t^j|R_k, s_i) F(R_k|s_i) = \sum_{k=i,-i} P(x_t^j|R_k) F(R_k|s_i) = \\ &= P(x_t^j|R_i) \frac{1-e^{-\frac{1}{\lambda}\tilde{K}L(R_{-i}|R_i)}}{1-e^{-\frac{1}{\lambda}(\tilde{K}L(R_2|R_1)+\tilde{K}L(R_1|R_2))}} + \\ &+ P(x_t^j|R_{-i}) \left(1-\frac{1-e^{-\frac{1}{\lambda}\tilde{K}L(R_{-i}|R_i)}}{1-e^{-\frac{1}{\lambda}(\tilde{K}L(R_2|R_1)+\tilde{K}L(R_1|R_2))}}\right), \quad i=1,2. \end{aligned}$$

The first equality follows by the Law of Total Probability. The second is implied by $P(x_t^j|R_k, s_i) = P(x_t^j|R_k)$, which in turn follows from the fact that s_i is a signal about R and conditional on knowing R , s_i provides no information about x_t . The third equality follows by Proposition 6.2.

ii) follows jointly by i) and Corollary 6.2. $\square$

Theorem 4.1. *The following are true under the optimal information strategy,¹⁶*

i) *The unconditional probability of observing s_1 is given by*

$$p_{s\lambda} = p_0 \frac{1}{1-e^{-\frac{1}{\lambda}\tilde{K}L(R_2|R_1)}} + (1-p_0) \left(1-\frac{1}{1-e^{-\frac{1}{\lambda}\tilde{K}L(R_1|R_2)}}\right),$$

ii) *The posterior probability of R_i given $s_\lambda = s_i$ is given by*

$$F(R_i|s_i) = \frac{1-e^{-\frac{1}{\lambda}\tilde{K}L(R_{-i}|R_i)}}{1-e^{-\frac{1}{\lambda}(\tilde{K}L(R_2|R_1)+\tilde{K}L(R_1|R_2))}} \quad i=1,2$$

iii) *For each i , the responses of $F(R_i|s_i)$ to local changes in λ is described as follows:*

$$\frac{\partial}{\partial \lambda} (F(R_i|s_i)) > 0$$

¹⁶For high values of λ , the agent might find it optimal not to gather any information, i.e. he might select completely uninformative signal function that always returns one value. To keep fidelity with reality, I assume throughout the paper that the prior beliefs, the costs of mistakes and the value of λ are such that the unconditional distribution of s_λ is non-degenerate.

iff

$$\tilde{K}L(R_i||R_{-i})F(R_i, s_{-i}) > \tilde{K}L(R_{-i}||R_i)F(R_{-i}, s_{-i}),$$

iv) The optimal information strategy is perfectly informative in the limit,¹⁷ i.e. as $\lambda \rightarrow 0$,

$$p_{s_\lambda} \rightarrow p_0$$

and

$$F(R_i|s_i), F(s_i|R_i) \rightarrow 1,$$

v) The posterior probability of x_t^j given $s_\lambda = s_i$ is given by

$$P(x_t^j|s_i) = P(x_t^j|R_i)F(R_i|s_i) + P(x_t^j|R_{-i})F(R_{-i}|s_i), \quad i = 1, 2$$

and

$$P(x_t^j|s_i) \xrightarrow{\lambda \rightarrow 0} P(x_t^j|R_i).$$

Proof. Parts i), ii) and v) are proved as Propositions 6.1, 6.2 and 6.3, respectively. Part iii) is proved as Corollary 6.3 and Part iv) - as Corollaries 6.1 and 6.2. $\square$

Equilibrium: Solution

Theorem 5.1. Let $P_{Z,0}$ denote the time-zero observed price of an asset with pay-off function $Z(x_t)$ following a signal realization s_i and when the cost of information is λ . Let $P_{Z,0}^i$ denote the price of the asset in regime R_i if it were fully observable, and $F(R_i|s_i)$ - subjective posterior probability of R_i after observing s_i under optimal information strategy. Then,

$$P_{Z,0} = F(R_i|s_i)P_{Z,0}^i + F(R_{-i}|s_i)P_{Z,0}^{-i}.$$

Proof. Note first that by the no-arbitrage principle,

$$\begin{aligned} P_{Z,0} &= \sum_{x_t} q(x_t) Z(x_t) = \\ &= \sum_{x_t} P(x_t|s_\lambda = s_i) \frac{q(x_t)}{P(x_t|s_\lambda = s_i)} Z(x_t) = E_{P(x_t|s_i)} \left[\frac{q(x_t)}{P(x_t|s_\lambda = s_i)} Z(x_t) \right], \end{aligned}$$

¹⁷ The claim rests on the assumption that the prior beliefs about R are unbiased.

where $E_{P(\cdot|s_i)}[\cdot]$ denotes the expectation induced by posterior beliefs $P(x_t|s_\lambda = s_i)$. By the FOC of the portfolio selection problem under beliefs $P(x_t|s_\lambda = s_i)$ (see (6.1)),

$$\frac{q(x_t)}{P(x_t|s_\lambda = s_i)} = \beta^t \frac{c(x_t)}{c(x_0)}$$

and by market clearing,

$$c(x_t) = \omega(x_t) \quad \forall x_t.$$

Hence,

$$P_{Z,0} = E_{P(\cdot|s_i)} \left[\beta^t \frac{\omega(x_t)}{\omega(x_0)} Z(x_t) \right].$$

Finally, by Part v) of Theorem 4.1,

$$\begin{aligned} P(x_t^j|s_\lambda = s_i) &= P(x_t^j|R_i) F(R_i|s_i) + \\ &+ P(x_t^j|R_{-i}) F(R_{-i}|s_i) \end{aligned}$$

and so

$$\begin{aligned} P_{Z,0} &= E_{P(\cdot|s_i)} \left[\beta^t \frac{\omega(x_t)}{\omega(x_0)} Z(x_t) \right] = \sum_{x_t} \beta^t \frac{\omega(x_t)}{\omega(x_0)} Z(x_t) P(x_t|s_\lambda = s_i) = \\ &= \sum_{x_t} \beta^t \frac{\omega(x_t)}{\omega(x_0)} Z(x_t) P(x_t|R_i) F(R_i|s_i) + \\ &+ \sum_{x_t} \beta^t \frac{\omega(x_t)}{\omega(x_0)} Z(x_t) P(x_t|R_{-i}) F(R_{-i}|s_i) = \\ &= F(R_i|s_i) P_{Z,0}^i + F(R_{-i}|s_i) P_{Z,0}^{-i}. \end{aligned}$$

□